

Fixed Oils, Fats and Waxes

Fixed Oils and Fats

- Fixed oils and fats are heterogeneous mixtures of different esters of glycerol with higher long chain fatty acids (4-24 carbon atoms)
 - Fixed oils and fats differ is in the proportion of saturated fatty acids occurring in their triglycerides
- Fixed oils have higher proportion of unsaturated fatty acids in their triglycerides while fats have higher proportion of saturated fatty acids in their triglycerides
 - The complete hydrolysis fixed oils and fats yield glycerol and mixtures of fatty acids
 - In plants fixed oils or fats occur in seeds

Physical Properties of Fixed Oils and Fats

- At room temperature, fixed oils are liquids and fats are solid or semisolid
- Greasy substances
- have lower specific gravity than water
- Insoluble in water and alcohol
- Soluble in non-polar solvents such as petroleum ether, ether, hexane, chloroform, benzene etc.
- non volatile
- Melting points and boiling points are not usually sharp.
- Pure fats and oils are colorless and odorless
- Color and odor is always due to plant pigments or contaminants

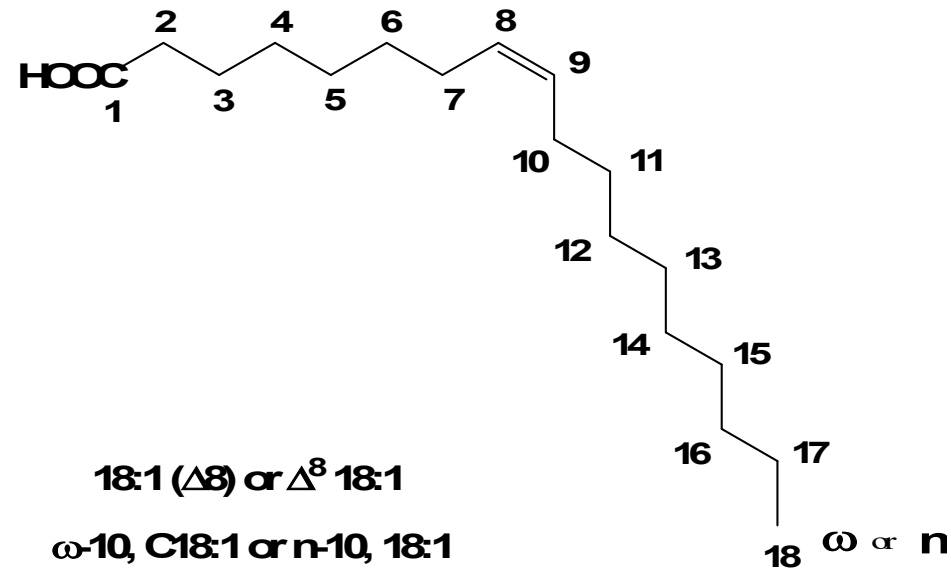
Fatty Acids Found in Fixed Oils and Fats

- In fixed oils and fats, fatty acids do not normally occur free in significant amount
- They are obtained by hydrolyzing the triglycerides of fixed oils or fats
- They have a single carboxyl group and a long non-polar hydrocarbon tail
- Naturally occurring fatty acids differ from one another
 - by their chain length
 - in the presence, number and position of their double bonds
 - in the presence or absence of alkyl branches

- Nearly all fatty acids in nature have an even number of carbon atoms
 - According to chain length FAs can be
 - short chain FA: 2-4 carbon atoms
 - medium chain FA: 6 –10 carbon atoms
 - long chain FA: 12 – 24 carbon atoms
- Fatty acids with 16 and 18 carbons are the most abundant

- In nature, unsaturated fatty acids are **twice as abundant** as saturated fatty acids
- Double bond is always ***cis*** in natural fatty acids
 - This lowers the melting point due to “kink” in the chain
- Two or more double bonds exist as non-conjugated double bonds
- Most polyunsaturated fatty acids are ω -6 fatty acids
- ω -3 fatty acids are found mainly in fish and fish products. They also occur in plant sources

- Longer chain and saturation increases melting point of fatty acids



Extraction

- From Plants

- **Expression method:** In expression method the seeds of the oil-bearing plant are crushed between two heavy rollers to squeeze the oil out
- If the heavy rollers are heated the method is called **hot expression**
 - Hot expression **increases yield** but it also increases level of contamination, thus lower quality
- If the heavy rollers are not heated the method is called **cold expression**
 - Cold expression is usually associated with **low yield** but gives good quality oil with low level of contamination

- **Solvent extraction:** could be used for seeds
 - It gives good yield but poor quality products at the same time

From Animals

- By heating the animal tissue with steam, the fat is melted and separated from the remaining matter.
- Alternatively adipose tissues are cut and mixed with hot water to melt the pure fat which is then separated by decanting.

Refining Process for Fixed Oils and Fats

Degumming:

- This is the first step in the refining process of many oils
- It is carried out to remove phosphatides (phospholipids)
- The oil is mixed with water to hydrate the phosphatides, which are then removed by centrifugation
- Phosphoric acid, citric acid or silica gel are added to enhance the process

Alkali refining:

- The degummed oil is then treated with alkali to remove free fatty acids
- The free fatty acids will form a soap that can be removed by centrifugation
- Any residual soap is removed by hot water washing

Bleaching:

- Trace metals, pigments (e.g. chlorophyll) and oxidation products are removed using bleaching clay like **Acid activated bentonites**, which adsorb the impurities
- Bleached oils are nearly colorless

Winterization:

- All medicinal oil and veterinary oils are cooled to about 0°C, which causes **stearin** (triglycerides with a higher saturated fatty acid content like myristic acid, palmitic acid and stearic acid) to solidify
- Some oils are thus chilled at 0°C with gentle agitation to precipitate high melting fraction
- The fraction which settles out is called stearin

Deodorization:

- Deodorization is a steam distillation process carried out in vacuum to remove volatile compounds from the oil

Commercial Applications of Fixed Oils and Fat

- As food
- Pharmaceutical excipients
- Cosmetics
- Lubricants
- Soap production
- In paint and varnish manufacturing (drying oils)

Examples of Fixed Oils and Fats Used in Pharmacy and Medicine

Peanut oil (arachis oil)

- The seed of the cultivated varieties of *Arachis hypogaea* (Fabaceae)
- cold extraction followed by solvent extraction of the cake
 - two oil extracts are mixed before purification
- is a colorless or a pale-yellowish liquid



Fatty Acid Composition

- 8-20% palmitic acid (16:0)
- 1-4% stearic acid (18:0)
- **56-86% oleic acid (18:1)**
- 4-20% linoleic acid (18:2)
- Minor fatty acids are:
 - arachidic acid (20:0),
 - behenic acid (22:0)
 - lignoceric acid (24:0)

Use

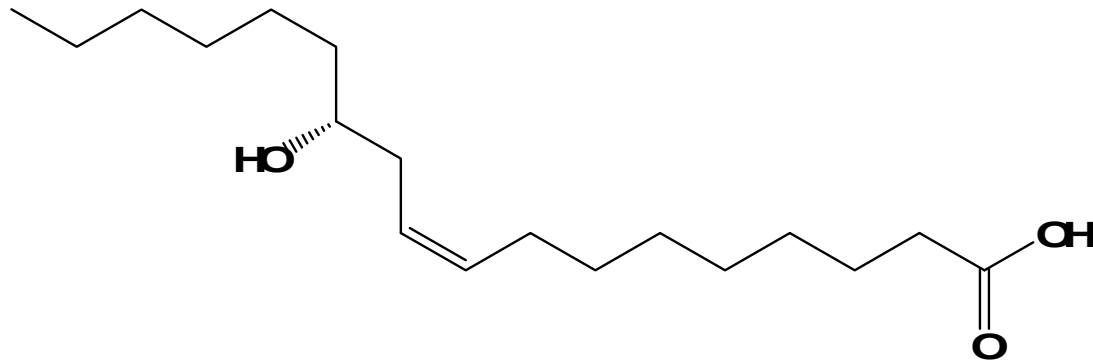
- As a vehicle for intramuscular injections
- preparation of margarine
- Note:
 - Because of its similarity with olive oil, peanut oil is commonly used to adulterate olive oil
- may cause serious allergic reaction

Castor Oil

- cold expression from the ripe seeds of *Ricinus communis* (Euphorbiaceae)
- After extraction, castor oil has to be steamed, filtered and bleached
 - Steaming is required, in order to denature ricin
 - an extremely toxic glycoprotein (lectin)
- medicinal castor oil is colorless or pale yellow liquid of extremely high viscosity

Fatty Acid Composition

- **90% ricinoleic acid**
- 1% isoricinoleic acid
- (3%) oleic acid
- (3-4%) linoleic acid, dihydroxystearic acid, palmitic acid and stearic acid



Uses

- The oil is stimulant **purgatives**
 - This is due to ricinoleic and isoricinoleic acids which are liberated from their triacylglycerol forms by pancreatic lipase in the duodenum
 - act on the walls of the small intestine
 - onset of action is thus delayed by 1-2 hrs
 - It is applied in food poisoning and before intestinal examination

- Externally as emollient.
- Because of its high viscosity, it is used as stiffening agent in ointments and creams.
- Polyethoxylated castor oil is used in some intravenous preparations, which contain drugs of low water solubility.
- Also used for the preparation of lubricants

Remark

- Note that the cake obtained after the extraction of oil is not fit for use as cattle food
 - because it contains high amount of ricins
 - the cake could be used as chicken food
- Children should not be allowed to play with castor beans



Almond Oil

- cold expression from the seed kernels of *Prunus amygdalus* var. *dulcis* (sweet almond) or *Prunus amygdalus* var. *amara* (fam: Rosaceae).
- Almond oil is pale yellow liquid

Fatty acid composition

- 4-9% palmitic acid (16:0)
- 1% stearic acid (18:0)
- **62-86% oleic acid (18:1)**
- 20-30% linoleic acid (18:2)



Use

- In cosmetology and dermatology
 - Relieve skin irritation and sensation of heat
- Could also be used as a vehicle for oily injections

Remark:

- HCN may occur in bitter almond oil and therefore its presence and level should be determined

Linseed Oil (Flaxseed Oil)

- by hot expression from the dried, ripe seeds of *Linum usitatissimum* (Linaceae)
- is a yellow liquid when fresh but when exposed to air it gradually thickens and darkens in color and form a hard varnish



Fatty Acid Composition

- **36-50% α -linolenic acid (18:3 ($\Delta^{9,12,15}$))**
- 23-24% linoleic acid (18:2)
- 10-18% oleic acid (18:1)
- 7% stearic acid (18:0)
- 6% palmitic acid (16:0)

Uses

- Linseed oil is not suitable for food because of its disagreeable taste
- Because of its' high content of unsaturated fatty acid composition
 - it is a drying oil and thus important in paint and varnish production
- Linseed cake is valuable cattle food

- After hydrolysis it is very effective antibacterial agent and is used for wound dressing
 - active against *staphylococcus aureus* strains resistant to antibiotics
- reduces both the serum cholesterol level and blood pressure
- ω -3 FAs inhibit formation of thromboxane A_2 , which is required for platelet aggregation and clot formation
- exerts **anti-inflammatory** activity so help in rheumatoid arthritis and gastrointestinal inflammation
- It also alleviates menstrual and postmenopausal problems

Sesame Oil

- from the seeds of *Sesamum indicum* (Pedaliaceae) by expression or extraction
 - Fatty acid composition
 - 7-12% palmitic acid (16:0)
 - 3-6% stearic acid (18:0)
 - **35-50% oleic acid (18:1)**
 - **35-50% linoleic acid (18:2)**
 - The fixed oil also contains a considerable mounts of plant sterols, tocopherols (vitamin E).
 - Uses:
 - As mild laxative
 - In cosmetics
 - As vehicle for drugs
 - Sesame oil is commonly used to adulterate other oils



Theobroma Oil (Cocoa Butter)

- is the fat obtained by hot expression from the roasted kernels of *Theobroma cacao* (Sterculiaceae)
- is yellowish white solid
- has a relatively sharp melting point: 30-35°C



Fatty Acid Composition

- **25-29% palmitic acid (16:0)**
- **32-37% stearic acid (18:0)**
- **34-36% oleic acid (18:1)**
- 2% linoleic acid (18:2)

Uses

- for the preparation of suppositories and lipsticks
- most expensive of the commercial fixed oils.
- may be adulterated with waxes, stearin, animal and vegetable fats

Cotton Seed Oil

- hot expression from the seeds of various *Gossypium* (Malvaceae).
- High temperature is employed in the extraction of cotton seed oil, in order to inactivate gossypol, which can cause sterility in males
- Gossypol is used as male contraceptive in China
- Cotton seed is a pale-yellow, semi-drying oily liquid

Fatty Acid Composition

- 1-3% myristic acid (14:0)
- **18-45% palmitic acid (16:0)**
- 1-9% stearic acid (18:0)
- 10-32% oleic acid (18:1)
- **31-53% linoleic acid (18:2)**

Use:

- as vehicle for injections
- After hydrogenation, as substitute for lard.



Lard

- is the purified internal fat of the hog (*Sus scrofa*: Suidae)
- is a soft, white fat with non-rancid taste
- Its melting point is from 34-41°C
- The fatty acids occurring in lard are:
 - mainly (65%) saturated ones such as myristic acid, palmitic acid, stearic acid
 - about 35% unsaturated ones such as oleic acid
- can easily get rancid and benzoin is added as a preservative
- is used as ointment base



Code Liver Oil

- prepared from the fresh liver of the cod (*Gadus callarias*: Gadidae).
 - The cod livers which contains about 50% oil, is removed immediately after the fish are caught
 - The livers are then transferred to a steamer where the oil will be liquefied by the high temperature and separates
 - The oil is rapidly heated to 77 °C
 - Various reagents are added
 - removes impurities and
 - cause the dissolution of liver tissue fragments
 - The resulting oil is centrifuged
 - The above process is repeated twice and the oil is water-washed



- Drying is done in vacuum drying towers
- Winterization to remove stearin
- Deodorization
 - Is carried out by steaming the oil under vacuum to remove aldehydic and ketonic impurities, which otherwise would impart disagreeable odor
- Standardization
 - After blending, the vitamin A and vitamin D content of the oil is determined
 - In BP, for instance, 1 gram of cod liver oil is required to have 600 Units of vitamin A and 60 Units of vitamin D

- it is not allowed to add any artificial vitamin A and vitamin D into cod liver oil. Vitamin E, however, could be added
- Cod liver oil is composed of glycerides of :
- ω -3 series
- unsaturated fatty acids (85%) and saturated fatty acids (15%)
 - Besides the common fatty acids, ω -3 series fatty acids such as eicosapentaenoic acid 20:5 ($\Delta^{5, 8, 11, 14, 17}$) and decosahexaenoic acid 22:6 ($\Delta^{4, 7, 10, 13, 16, 19}$)

Uses:

- As source of vitamin A and vitamin D



WAXES

- Waxes are mixtures of long chain usually saturated fatty acids (C_{14} - C_{26}), long chain alcohols (C_{18} - C_{36}) and their simple esters
- Waxes are solid at room temperature but soften and melt with heat
- Waxes are abundant in plants leaves' cuticle layer.
- commercially they are obtained from very few sources
- Animal and insect waxes are also used

Beeswax (yellow wax)

- is the purified wax from the honey comb of bee (*Apis mellifera*: Apidae).
- Honeycomb, after separation from the honey, is melted in water, then cooled and remelted and finally strained and allowed to harden in mold



Honeycomb is composed of

- myricyl palmitate $C_{15}H_{31}COOC_{30}H_{61}$
- myricyl cerotate $C_{25}H_{51}COOC_{30}H_{61}$
- similar high molecular weight esters



- yellow wax is used as
 - as stiffening agent
 - as base in plasters
 - in the manufacture of candles, cosmetics and polishes.
- White wax (bleached beeswax)
 - White wax is bleached yellow wax.
 - Bleaching is carried out by repeated exposure to sunlight
 - White wax is employed in ointments and cold creams

Spermaceti

- is a waxy substance obtained from the head of the sperm whale (*Physeter macrocephalus*: Physeteridae)



- valued for its quality emollient property and commonly added in cold creams
- sperm whale has become an endangered species
 - natural spermaceti is no more available
- Instead, synthetic spermaceti (cetyl esters wax) is used
 - Synthetic spermaceti is a mixture consisting primarily of esters of saturated fatty alcohols (C_{14} to C_{16}) and saturated fatty acids (C_{14} to C_{18})

Volatile oils



Volatile oils

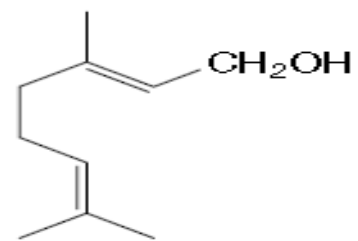
- Volatile or **essential oils**, as their name implies, are volatile in steam.
- They differ entirely in both chemical and physical properties from fixed oils.
- They are secreted in oil cells, in secretion ducts or cavities or in glandular hairs.
- They are frequently associated with other substances such as gums and resins and themselves tend to resinify on exposure to air
- The global essential oils market is expected to grow from \$9.62 billion in 2021 to \$18.25 billion in 2028 at a CAGR of 9.57% in forecast period, 2021-2028.

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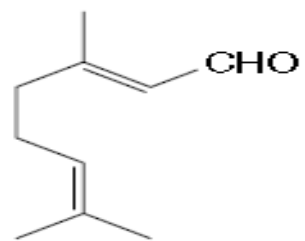
- Volatile oils are used for their food, cosmetics, pharmaceutical industry & for therapeutic action
 - Flavouring (e.g. oil of lemon)
 - perfumery (e.g. oil of rose)
 - Therapeutic administered as inhalations (e.g. eucalyptus oil),
 - Orally (e.g. peppermint oil)
 - Gargles and mouthwashes (e.g. thymol)
 - Transdermally (many essential oils including those of lavender, rosemary and bergamot for aromatherapy).

Composition of volatile oils

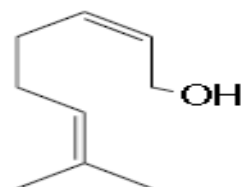
- Volatile oils are generally mixtures of **hydrocarbons** and **oxygenated** compounds derived from these hydrocarbons.
 - Oxygenated hydrocarbons are most abundant and are responsible for the odor and taste
- Many oils are terpenoid in origin; but some like cinnamon and clove contain principally aromatic (benzene) derivatives mixed with the terpenes.
- A few compounds (e.g. thymol, carvacrol), although aromatic in structure, are terpenoid in origin.
- essential oils should be stored in well-filled, airtight containers, protected from light and heat.



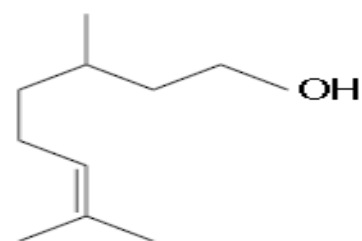
Geraniol



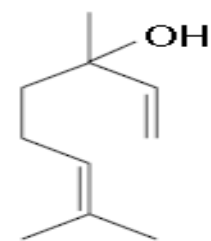
Citral



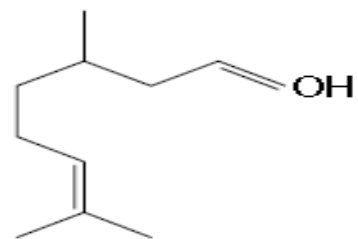
Nerol



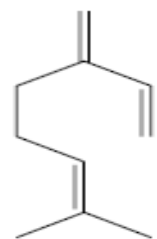
Citronellol



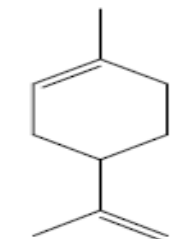
Linalool



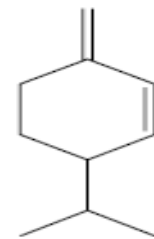
Citronellal



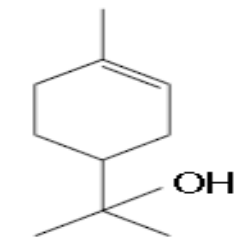
β -Myrcene



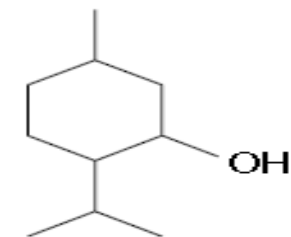
Limonene



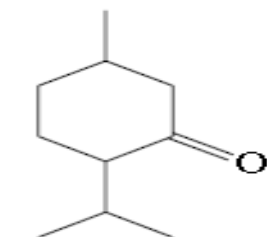
β -Phellandrene



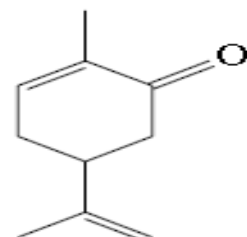
α -Terpineol



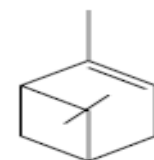
Menthol



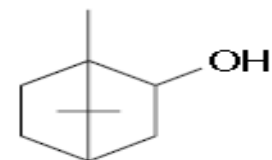
Menthone



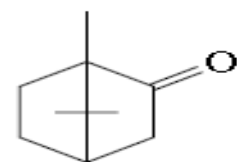
Carvone



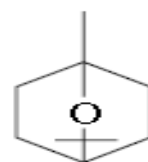
α -Pinene



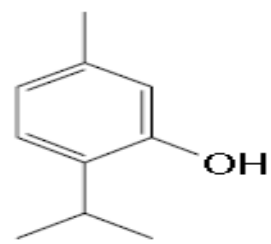
Borneol



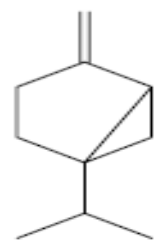
Camphor



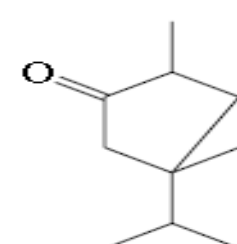
Cineole



Thymol



Sabinene



Thujone

- Type of Material and volatile oils
 - Dried Bud -Cloves
 - Flowers- Jasmin
 - Herbaceous Plant-Sage
 - Twigs Bark-Cinnamon
 - Wood Sandal - wood
 - Rhizome -Ginger
 - Fresh Fruit -Lime
 - Dried Fruit - Fennel
 - Animal - Castoreum

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Evaluation

- Various pharmacopoeial procedures are given for the evaluation of volatile oils.
- Odour and taste are obviously important in the preliminary examination;
 - however oils should be tasted only after dilution with a sugar solution in ethanol as prescribed in the *BP*.
- Physical measurements including optical rotation, relative density and refractive index are regularly employed for identification.
- TLC with vanillin spray to give purple spots
- GC-MS is the golden standards for evaluation

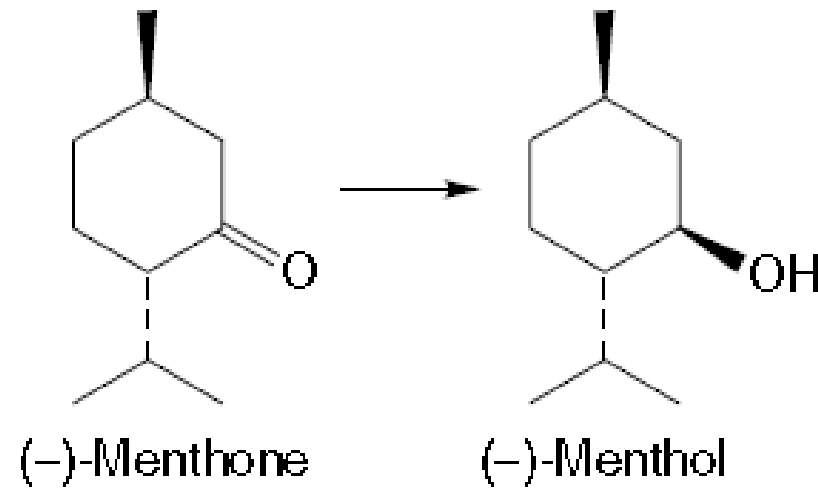
Extraction

- **Distillation:**
 - Hydro distillation
 - Steam distillation
- **Solvent extraction:**
 - high yield but low quality
- **Microwave extraction :**
 - The sudden increase in temperature causes the cell walls of the essential oil glands to rupture and release their oil to the surrounding solvent

- **supercritical liquid extraction (subcritical liquid CO₂)**
 - odorless, colorless, tasteless and non-toxic
 - non-combustible inexpensive and readily available easily removed leaving no solvent residue
 - because of low viscosity it can readily penetrate comminuted dry plant material
 - by varying the temperature and pressure it can be used in a more selective manner.

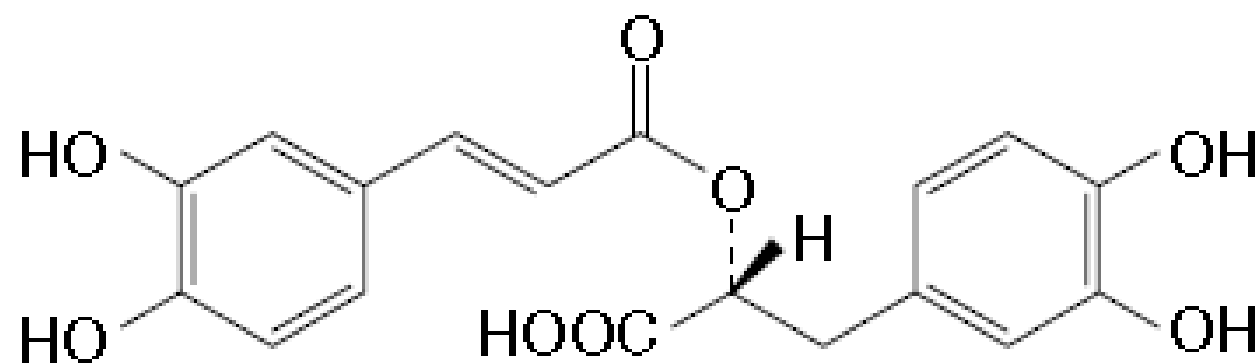
Oil of Peppermint

- 44% of free alcohols calculated as menthol and 15–32% of ketones calculated as menthone.
- Has carminative effect



Rosemary Oil

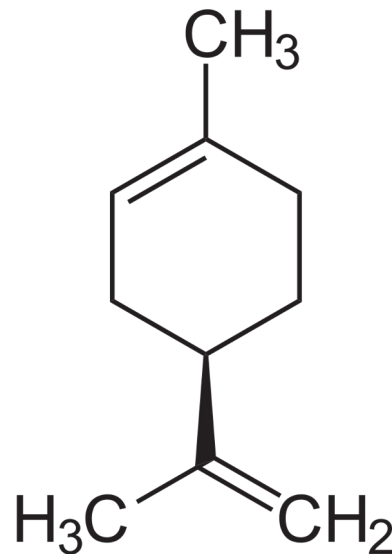
- Rosemary oil is steam distilled from the flowering aerial parts of *Rosmarinus officinalis* L.
- The fresh material yields about 1–2% of a colourless to pale yellow volatile oil with a very characteristic odour.
- The oil is frequently used in aromatherapy, in the perfumery industry and for the preparation of spirits and liniments for medical use;
 - has antibacterial and antispasmodic properties.
- The principal constituents are:
 - 1,8-cineole, borneol, camphor, bornyl acetate and monoterpene hydrocarbons, principally α -pinene and camphene.



Rosmarinic acid

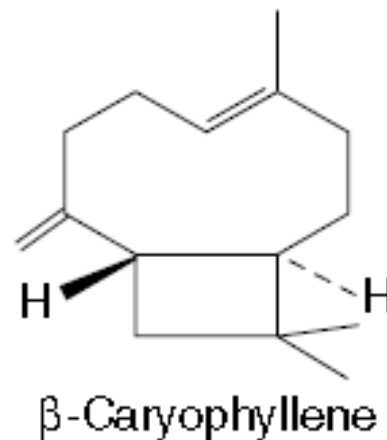
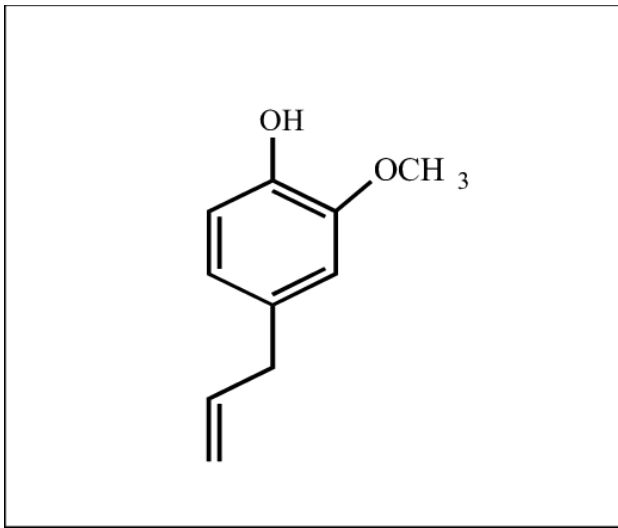
Orange Oils

- The volatile oil from the orange may be extracted by methods other than by distillation: **expression**
- These oils contain the terpene (+)-limonene and smaller quantities of citral, citronellal, methyl anthranilate
- **Used as flavoring in pharmaceutical industr**



Clove oil

- Cloves contain about 14–21% of volatile oil (see below),
 - Clove oil contains 84–95% of phenols (**eugenol** with about 3% of acetyleneugenol), sesquiterpenes (α - and β -caryophyllenes) and small quantities of esters, ketones and alcohols.
- It is used as a flavouring agent, stimulant, aromatic and antiseptic



Eucalyptus Oil

- Pharmaceutical Suitable oils are derived from *Eucalyptus. polybractea*, *E. smithii*, *E. globulus* and *E. australiana*.
- The chief requirements are a high cineole content and the absence of appreciable quantities of phellandrene and aldehydes
- ‘Citron-scented’ eucalyptus oil, which is derived from *E. citriodora*, is used in perfumery and contains a high proportion of the aldehyde citronellal.
- Oil of eucalyptus is a colourless or pale yellow liquid
- It is required to contain not less than 70.0% of cineole. 1,8-Cineole and o-cresol form a solid complex and the crystallizing

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- Eucalyptus oil is much used for alleviating the symptoms of:
 - nasopharyngeal infections, for treating coughs and as a decongestant.
 - It is taken internally in the form of mixtures, inhalations, lozenges and pastilles
 - applied externally as ointments and liniments.

